

How To Become a Physics Pro

Proposed by:

Team Curiosity

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Problem Statement

Let's be for real: Physics can be one of the most challenging subjects for Grade 8 students. Don't deny it, many students struggle to understand concepts such as *motion, force, speed, work, energy, gravitational laws, and heat transfer* because the topics are way too complicated and require the strongest abstract and critical reasoning.

Did you know: Students often rely on memorisation (stock-knowledge) rather than dedicated understanding. When dozens of formulas, units, and scientific terms accumulate, scholars lose confidence and stimulus. This results in low scores, frustration, and a weaker appreciation for how fun physics can be.

We interviewed **Katara, an 8th grader from 8-Camia**, who communicated that she finds physics “too heavy” and wishes for a study method that explains the concepts thoroughly, makes learning interactive, and helps her review without stress. Her very experiences are what grade 8 students face, and let's be real, that summative assessment exam for the grade 8's would really be an entire list piled up all together.

Create a software that can be a viable study method for students, whether they use a premade quiz for a topic or they make their own quiz to help others! Students often struggle to find effective and interactive study methods tailored to their individual learning styles and their own academic needs. Pressure builds up when the entire school year's topic of numerous subjects is going to be the final decision maker of whether you will be a Director's Lister.

Through this project, we can create an engaging way of studying or a collaborative way of learning for Physics. This is the project “**Gravity pulls with Curiosity.**” Where limited ideas are taken into software with no limits.

Project Objectives

Pressure builds up: When faced with an exam that shows your efforts, you don’t want to fail it, duh! I, Edgar, am a Physics geek. I want to show my peers how Physics can be an engaging subject, for them to understand why I cherish it so much. I, Viktoria, am a compassionate person. I want to embody kindness and to show it; this project is for me. I, Seth Bito, a lover man, with loyalty, we can show that challenges are a step back. This is how we can give back to our peers, to firstly eliminate their fears.

But let’s be honest: Memorising thousands of terms can feel overwhelming, it turns us into a dictionary that doesn’t even have definitions, which is why

This project aims to:

1. Provide a platform for **Physics** quizzes focused on Grade 8 topics, such as:

- Ideas of Motion
- Kinematic Quantities
- Newton’s Law of Motion
- Force
- Impulse and Momentum
- Conservation of Momentum
- Historical Development of the Universal Law of Gravitation
- Gravitational Force, Field, and Potential Energy
- Acceleration due to Gravity
- Conservation of Mechanical Energy
- Heat Transfer
- Consequences of Heat Transfer
- Climate Change and Renewable Energy
- Introduction to Waves

- Characteristic Behaviors of Waves
 - Sound as a Wave and the Physics of Echolocation
 - Historical Development of Ideas about Light
 - Colors
 - Wave Nature of Light
 - Electromagnetic Wave
 - Electricity and Magnetism – Electric Charge and Charging Methods
 - Problem Solving in Coulomb's Law
 - Electricity and Magnetism – Voltage, Current, and Resistance
 - Electromagnetism
 - Applications of Electricity and Magnetism/ St. Elmo's Fire
2. **Improve students' comprehension** by giving feedback, hints, and corrections in real time.
- Offer clear and well-structured explanations on the validity of the answer; this is why you were correct/wrong.
3. Make learners appreciate physics by showing that it can be less intimidating and easier to understand.

Actual Inputs and Outputs

User Log-In System

Inputs:

- Select option:
 - [1] Log In
 - [2] Sign Up
- Enter username
- Enter password

Outputs:

- "Login Successful"
- "Account Created Successfully"
- "Invalid Credentials. Please try again."

User Interface

System Outputs:

- Loading screen with animation
- Banner display: *"Welcome to Gravity Pulls With Curiosity"*

After the authentication, the user is presented with the main menu:

Main Menu Options (Inputs):

- [1] Physics Mission
- [2] Instructions Manual
- [3] Exit

System Responses (Outputs):

- Navigation to selected feature
- System notifications (e.g., quiz ready, instructions displayed)

If the User Selects “Physics Mission”

Inputs:

1. **Select Quarter / Topic:**
 - Quarter 1
 - Quarter 2
 - Quarter 3
 - Quarter 4
2. **Select Difficulty Level:**
 - Easy → 5 questions
 - Medium → 10 questions
 - Hard → 20 questions
 - Limitless → All available questions
3. **Answer Input:**
 - Multiple Choice: A, B, C, D
 - True or False

Outputs:

For each question, the system displays:

- Topic
- Question text
- Answer choices

System Feedback:

- **If Correct:**
 - Provides explanation of the correct answer
 - Awards corresponding points
- **If Incorrect:**
 - Displays error message
 - Provides hint or explanation
 - Allows retry with a penalty of -0.25 points

True or False Questions

- User inputs **True** or **False**
- System provides feedback and explanation regardless of correctness
- Retry system applies similar to multiple-choice questions

Review Page

Outputs after quiz completion:

- Final score
- Percentage result
- List of incorrect answers
- Correct answers with explanations

This feature helps users focus on areas that need improvement or the questions they got wrong.

If the User Selects “Instructions”

Outputs:

PROGRAM INSTRUCTIONS & MANUAL

Description:

“Gravity Pulls With Curiosity” is a study tool designed for Grade 8 students to learn Physics....”

Instructions:

1. **Navigation:**
Use numbers [1–3] to select menu options, or type the menu options
2. **Answering Questions:**
Enter A, B, C, D, True, or False
3. **Scoring System:**
 - Correct answers earn full points
 - Incorrect answers deduct 0.25 points
 - Users may retry until correct

Menu Options:

- [1] Physics Mission → Start quiz
- [2] Instructions → View manual
- [3] Exit → Close program

Logic Plan

```
// ===== GLOBAL SETUP =====  
  
SET TeamMembers = ["Fortes, Edgar", "Bito, Seth", "Ancheta, Viktoria"]  
  
// ===== STARTUP =====  
  
CALL display_robot_watermark()  
CALL loading_bar_animation()  
  
// ===== AUTHENTICATION =====  
  
PRINT "Authentication Required"  
  
INPUT "Choose [1] Log In or [2] Sign Up" AS authorization  
  
IF authorization == "2" THEN  
    INPUT "Username" AS username  
    INPUT "Password" AS password  
    CALL update_data(username, password, signup=True)  
    PRINT "Account created successfully."  
ELSE IF authorization == "1" THEN  
    INPUT "Username" AS username  
    INPUT "Password" AS password  
    CALL get_user(username) AS account  
  
    IF account is valid AND password matches THEN  
        PRINT "Login successful."  
    ELSE  
        PRINT "Login failed."  
        TERMINATE PROGRAM  
    END IF  
END IF  
END IF
```

```
PRINT "Welcome to Team Curiosity!"

CLEAR SCREEN

// =====MAIN MENU =====

FUNCTION main_menu:

    PRINT "[1] Launch Physics Mission"

    PRINT "[2] Instructions"

    PRINT "[3] Exit"

    INPUT "Select option" AS choice

    RETURN choice

END FUNCTION


// ===== MAIN LOOP =====

WHILE TRUE:

    SET user_selection = CALL main_menu()

    IF user_selection == "1" THEN

        CALL physics_mission()

    ELSE IF user_selection == "2" THEN

        CALL show_instructions()

    ELSE IF user_selection == "3" THEN

        PRINT "Goodbye!"

        BREAK LOOP

    ELSE

        PRINT "Invalid choice. Try again."

    END IF

END WHILE
```

```
// ===== PHYSICS MISSION =====
```

```
FUNCTION physics_mission:
```

```
    PRINT "Select Quarter:"
```

```
    PRINT "[1] Q1 (Motion & Forces)"
```

```
    PRINT "[2] Q2 (Energy & Heat)"
```

```
    PRINT "[3] Q3 (Waves)"
```

```
    PRINT "[4] Q4 (Electricity & Magnetism)"
```

```
    INPUT sub_choice
```

```
    IF sub_choice == "1" THEN
```

```
        SET filename = "physics_q1.csv"
```

```
    ELSE IF sub_choice == "2" THEN
```

```
        SET filename = "physics_q2.csv"
```

```
    ELSE IF sub_choice == "3" THEN
```

```
        INPUT "Select Mode [1] MC or [2] TF" AS mode
```

```
        IF mode == "1" THEN
```

```
            SET filename = "physics_q3_mc.csv"
```

```
        ELSE
```

```
            SET filename = "physics_q3_tf.csv"
```

```
        END IF
```

```
    ELSE IF sub_choice == "4" THEN
```

```
        INPUT "Select Mode [1] MC or [2] TF" AS mode
```

```
        IF mode == "1" THEN
```

```
            SET filename = "physics_q4_mc.csv"
```

```
        ELSE
```

```
        SET filename = "physics_q4_tf.csv"
    END IF
END IF

OPEN filename → LOAD INTO questions_list
PRINT "Select Difficulty:"
PRINT "[1] Easy (5)"
PRINT "[2] Medium (10)"
PRINT "[3] Hard (20)"
PRINT "[4] Limitless"
INPUT difficulty_choice
IF difficulty_choice == "1" THEN
    SET limit = 5
ELSE IF difficulty_choice == "2" THEN
    SET limit = 10
ELSE IF difficulty_choice == "3" THEN
    SET limit = 20
ELSE
    SET limit = LENGTH(questions_list)
END IF

SHUFFLE questions_list
SET selected_questions = FIRST limit QUESTIONS
SET total_score = 0
FOR EACH question IN selected_questions:
    SET value = score based on difficulty
    SET current_score = value
    SET answered_correctly = FALSE
```

```

WHILE answered_correctly == FALSE AND current_score > 0:
    DISPLAY question, topic, and choices
    INPUT user_answer
    IF user_answer == correct_answer THEN
        PRINT "Correct!"
        PRINT explanation
        ADD current_score TO total_score
        SET answered_correctly = TRUE
    ELSE
        SUBTRACT 0.25 FROM current_score
        IF current_score > 0 THEN
            PRINT "Incorrect. Try again."
        ELSE
            PRINT "No points left."
            PRINT correct answer and explanation
        END IF
    END IF
END IF
END WHILE
END FOR
PRINT "Final Score:", total_score
CALL update_data(username, score=total_score)
END FUNCTION

```

```

// ===== INSTRUCTIONS =====

```

```

FUNCTION show_instructions:

```

```

    PRINT "Displays program manual and usage guide."

```

WAIT FOR USER INPUT
END FUNCTION

Features

Through this **software**, students can venture into these features:

- **Take quizzes:** In this feature, we used our provided SGs about Physics for an efficient way of making these premade quizzes.
 - If you get it **correctly**, you will get an explanation of why it is correct. This is to take your knowledge onto the spiritway of curiosity, or in simpler terms, to add to what you know.
 - If you get it **incorrectly**, you will get an explanation on why it is wrong. But it doesn't mean you can't take the question again, just like in Duolingo, the question comes back. If you get the question wrong, you will have a .25 deduction, so you can keep taking the question until it's correct! Scores can be an integer or a floating-point number.
- **Review Correct and Incorrect:** In this software, the user will be able to review what answers they got correctly and what answers they got wrong to focus on. It also shows why the answer was incorrect, what the correct answer is, and primarily explains!
- **Immediate Feedback**

Conclusion

In conclusion, “**Gravity pulls with Curiosity**” is an engaging software program designed to simplify and make learning easier by providing an interactive and collaborative platform for our fellow students to engage with various topics about physics through quizzes and other features. By allowing users to create and participate in quizzes, our platform aims to develop curiosity, create creativity, and exemplify critical thinking while promoting academic excellence. With features such as creating your own quizzes, taking premade quizzes, and collaborating with different people for a study session, testing the user’s knowledge. This software has the potential to make learning fun, accessible, and effective for students.

By empowering users to both create and participate, the platform actively works to foster **curiosity, spark creativity, and develop critical thinking skills**, all while encouraging academic excellence. Key features, including quiz creation, access to a library of premade content, and facilitated study collaboration, position this software to make learning **fun, accessible, and profoundly effective**.

“**Gravity pulls with Curiosity**” can help bridge the gap between knowledge and application, contributing to the development of a more informed, engaged, and curious community. Now our students can take that Summative Achievement Exam with ease, for Physics, relieving a big burden!

Team Curiosity!

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